

Clifford J. Crafford
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EDUCATION

BS in Mathematics with Statistics Concentration, <i>magna cum laude</i> University of Memphis, Memphis, TN	December 2017
MS in Applied Statistics University of Memphis, Memphis, TN	May 2021
PhD in Applied Statistics University of Memphis, Memphis, TN	May 2025

TEACHING EXPERIENCE

Graduate Teaching Assistant – NSF #182161 <i>University of Memphis, Math Department</i>	August 2019 – May 2020
• Tutoring students multiple times throughout the week in various math subjects. • Teaching courses and working as an instructor both inside and outside of the classroom, including the administering of tests. The course taught was Calculus 1, however, tutoring has also been provided for a variety of math courses. • Maintaining a schedule with office hours and appropriate contact measures to aid outside of classes.	

RESEARCH EXPERIENCE

Research Assistant – NSF #1918751 <i>University of Memphis, Statistics Department</i>	Feb 2020 – Present
• Assistant to Dr. Dale Bowman to work on the development of “teaching effective methods” for computer science and data science curriculums. • Delivered a presentation on the work done to develop a topics spreadsheet.	

Data Science and Applied Statistics Group
University of Memphis, Statistics Department

August 2019 – Present

- Working alongside professors to develop, submit, and/or publish papers to various journals and publications.

Summer Internship

July 2020 – September 2020

St. Jude Children's Research Hospital, Enterprise Analytics Group

- Worked alongside a research and analysis group led by Glenn Sigler, VP of Enterprise Analytics at St. Jude.
- Produced an independent report on the topic of “Impacts of Student Debt on Charitable Giving” and presented on it with an accompanying set of slides.
- Required weekly meetings, twice a week, to discuss progress and possible directions with the study and independent reading of various papers, studies, scholarly reports, and slight exploratory data analysis.

Summer Internship

June 2021 – August 2021

St. Jude Children's Research Hospital, Biostatistics

- Worked alongside a research and analysis group aiding Dr. Johnnie Bass and Dr. Deo Kumar Srivastava.
- Performed extensive analysis in R and SAS on data for the construction of tables and statistics relevant to the performance of a new scale to diagnose the severity of hearing loss in cancer treatment survivors.
- Required weekly meetings to discuss progress and possible directions with the study and reading of certain texts to perform the desired methods of analysis on categorical data and contingency tables. This work was performed in R Markdown and output as readable files as well as fully written and documented code that was future-proofed for further analysis once the internship expired.

Internship

January 2022 – August 2022

NASA-Marshall Space Flight Center, MSFC

- Worked alongside Dr. Enrique M. Jackson and other interns for the project “Computational Approaches to Understanding the Shape Memory of Ionic Polyimides – 21345”; UIN: 011072
- Attended meetings on the presentation and discussion of development for materials related to the Project Artemis mission.
- Developed slides and presentations regarding content relevant to the Project Artemis mission.
- Worked alongside Dr. Sridhar Gorti and other interns on the analysis of molecular images as part of Differential Dynamic Microscopy.

HONORS AND AWARDS

University of Memphis Dean's List for Academic Excellence

February 2014 – December 2017

GRANTS AWARDED

National Science Foundation #1918751. Advancing the Science of Learning Data Science with Adaptive Learning for Future Workforce Development. University of Memphis. January 2020 – Present

National Science Foundation #1821619. Collaborative Research: Promoting Success in Undergraduate Mathematics through Graduate Teaching Assistant Training. August 2019 – Present

TECHNICAL COMPETENCIES

R, RStudio, & RMarkdown

- Practical use of R for data analysis in a variety of data types ranging from time series analysis, variable simulation, and component analysis.
- Used for a variety of classes and in my Summer Internship with St. Jude to analyze some consumer data for its consideration into my report.
- Able to use RMarkdown for presenting results in a report format for viewing.

SAS

- Training in SAS for times series analysis, data analysis, and various applications towards other topics.

Python

- Able to use Python through Jupyter Notebook and Anaconda for programming.
- Used in Introductions to Machine Learning for building an Item-Item Collaborative Filtering program and further used to examine various programs for Machine Learning and Neural Networks.
- Utilized routinely in working on Kaggle machine learning and data science competitions.

C++

- Able to use C++ coding in Microsoft Visual Studio code environment.
- Used for various string editing projects not easily completed in R and R Studio.

IN-PROGRESS PUBLICATIONS

Bowman, D., E. George, X. Ju, C. Crafford, P. Ngutu. Shared Response Model. *Communications in Statistics – Theory and Methods*.

Bowman, D., C. Crafford, X. Ju, P. Ngutu. Inference for Exchangeable Multinomial Data with Unequal and Missing Cluster Sizes. *Communications in Statistics – Theory and Methods*.

C. Crafford, M. Salehabadi, E.O. George, K. Srivastava, K. Krull. Comparing Statistical Approaches for Analyzing Biomarkers Subject to Below Limit of Detection. *Biometrical Journal*.

ACADEMIC MEMBERSHIPS

American Mathematical Society (AMS)

December 2017 – Present

Data Science and Applied Statistics Group (DSAS)

August 2019 – Present

UNIVERSITY AND COMMUNITY SERVICE

Educational Support Program Learning Center – Mathematics Tutor

August 2019 – May 2020

LANGUAGES

English (Native)

Spanish (Able to read some and limited writing; Able to speak and understand, slowly)

Italian (Able to read and some writing; Able to speak and understand, slowly)

American Sign Language (Able to communicate with standard signs)

PROFESSIONAL REFERENCES

Lih-Yuan Deng, Ph.D.

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Glenn Sigler
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Johnnie Bass, Au.D, Ph.D
Research Audiologist, St. Jude Children's Research Hospital
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Enrique Jackson, Ph.D.

Adjunct Professor, Aerospace Polymeric Engineer for Materials Test Branch at NASA Marshall Space Flight Center

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